

Introduction to Multi-level Models

Introduction to multi-level models for correlated data

For more, see:

- [David Rocke's materials from the 2021 edition of this course](#)
 - May 25 - June 1 lectures
- Other UC Davis courses:
 - [EVE 225](#): “Linear Mixed Modeling in Ecology & Evolution”
 - * usually taught every other winter or spring by [Kate Laskowski](#)
 - * materials, including syllabus and lecture videos: <https://laskowskilab.faculty.ucdavis.edu/teaching-2/>
 - [STA/BST 224](#): “Analysis of Longitudinal Data”
 - * usually taught every spring by [Shuai Chen](#)
 - * should be accessible after completing Epi 204
 - [EPI 226](#) “Methods for Longitudinal & Repeated Measurement Data”
 - * usually taught by [Heejung Bang](#)
 - [PSC 205D](#) “Multilevel Models”
 - [PSC 205G](#) “Applied Longitudinal Data Analysis”
 - [STA 101](#) “Advanced Applied Statistics for the Biological Sciences”
 - [STA 207](#) “Statistical Methods for Research II”
 - [STA 232B](#) “Applied Statistics II”
 - * usually taught every winter by [Jiming Jiang](#)
 - [PLS 207](#): “Applied Statistical Modeling for the Environmental Sciences”
 - [EDU 236](#): “Application of Hierarchical Linear Models in Education Research”
 - [HDE 205](#): “Longitudinal Data Analysis”
- Books:
 - Dobson and Barnett (2018) [Chapter 11](#)

- Vittinghoff et al. (2012) [Chapter 7](#)
- Gelman and Hill (2007)
- Jiang and Nguyen (2021)
 - * by UC Davis Statistics Professor and GGE faculty member [Jiming Jiang](#)
- Faraway (2016)
- McCulloch et al. (2008)
- Hedeker and Gibbons (2006)
- Wakefield (2013)
- Zuur (2009)
- Diggle et al. (2013)
- Fitzmaurice et al. (2012)
- Fitzmaurice et al. (2009)
- Galecki and Burzykowski (2013)
- Congdon (2020)
- Molenberghs and Verbeke (2005)
- Verbeke and Molenberghs (2000)
- Jewell and Hubbard (2016)
 - * by UC Berkeley professors

Congdon, Peter D. 2020. *Bayesian Hierarchical Models: With Applications Using R, Second Edition*. 2nd edition. CRC Press.

Diggle, Peter, Scott Zeger, Patrick Heagerty, and Kung-Yee Liang. 2013. *Analysis of Longitudinal Data*. Second edition. Vol. 25. Oxford Statistical Science Series. Oxford University Press.

Dobson, Annette J, and Adrian G Barnett. 2018. *An Introduction to Generalized Linear Models*. 4th ed. CRC press. <https://doi.org/10.1201/9781315182780>.

Faraway, Julian J. 2016. *Extending the Linear Model with r: Generalized Linear, Mixed Effects and Nonparametric Regression Models*. 2nd ed. Chapman; Hall/CRC. <https://doi.org/10.1201/9781315382722>.

Fitzmaurice, Garrett M, Marie Davidian, Geert Verbeke, and Geert Molenberghs. 2009. *Longitudinal Data Analysis*. Chapman & Hall/CRC Handbooks of Modern Statistical Methods. CRC Press. <https://doi.org/10.1201/9781420011579>.

Fitzmaurice, Garrett M, Nan M Laird, and James H Ware. 2012. *Applied Longitudinal Analysis*. 2nd ed. Vol. 998. Wiley Series in Probability and Statistics. Wiley. <https://doi.org/10.1002/9781119513469>.

Galecki, Andrzej T., and Tomasz. Burzykowski. 2013. *Linear Mixed-Effects Models Using R : A*

- Step-by-Step Approach*. Springer Texts in Statistics. Springer. <https://doi.org/10.1007/978-1-4614-3900-4>.
- Gelman, Andrew, and Jennifer Hill. 2007. *Data Analysis Using Regression and Multilevel/Hierarchical Models*. Analytical Methods for Social Research. Cambridge University Press.
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- Jewell, Nicholas P, and Alan E Hubbard. 2016. *Analysis of Longitudinal Studies in Epidemiology*. Chapman & Hall/CRC Texts in Statistical Science. Taylor & Francis. <https://books.google.com/books?id=-LoLPQAACAAJ>.
- Jiang, Jiming, and Thuan Nguyen. 2021. *Linear and Generalized Linear Mixed Models and Their Applications*. Second edition. Springer Series in Statistics. Springer. <https://doi.org/10.1007/978-1-0716-1282-8>.
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- Molenberghs, Geert., and Geert. Verbeke. 2005. *Models for Discrete Longitudinal Data*. Springer Series in Statistics. Springer Science+Business Media, Inc. <https://doi.org/10.1007/0-387-28980-1>.
- Verbeke, Geert, and Geert Molenberghs. 2000. *Linear Mixed Models for Longitudinal Data*. 1st ed. Springer Series in Statistics. SpringerLink (Online service); Springer. <https://doi.org/10.1007/978-1-4419-0300-6>.
- Vittinghoff, Eric, David V Glidden, Stephen C Shiboski, and Charles E McCulloch. 2012. *Regression Methods in Biostatistics: Linear, Logistic, Survival, and Repeated Measures Models*. 2nd ed. Springer. <https://doi.org/10.1007/978-1-4614-1353-0>.
- Wakefield, Jon. 2013. *Bayesian and Frequentist Regression Methods*. 1st ed. 2013. Springer Series in Statistics. Springer New York.
- Zuur, Alain F. 2009. *Mixed Effects Models and Extensions in Ecology with r*. Statistics for Biology and Health. Springer.